

Transformer MLP Gate Thresholds Are Couplings to a Carried Reference Direction

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Abstract

The corpus-mean direction of a transformer’s residual stream is a component shared across all inputs, and is commonly removed by mean-centering before representational analysis on the assumption that it cannot carry input-specific signal. We present evidence that it is a functional component: the reference against which the MLP gate population sets its operating point. In Phi-2, an exact decomposition of resting gate pre-activations shows that the resting inhibition of $>99.9\%$ of gates is carried by the coupling $w \cdot b_L$ to the carried mean direction, at $50\text{--}60\times$ the explicit bias parameter, and the coupling is direction-specific: a random direction at matched norm orders the population’s firing rates at $\rho \leq 0.09$ (median over 50 draws; up to 0.22 in the tail) where the reference reaches 0.95. (That 0.95 on its own is near-tautological, since the resting term *is* the mean pre-activation; we derive its null rather than quote it as evidence.) Causally, removing the stream’s projection on the reference multiplies above-threshold firing by $\sim 9\times$, dose-monotonically, at $30\text{--}40\times$ a norm-matched control; a random-initialised twin is flat, and replacement tests show the function is carried by the direction, not the magnitude. A direction-matched control makes the same point: noise injected along the reference costs $8\text{--}42\times$ the same energy along a random direction. The decomposition replicates on four further families spanning both gate types (GELU with an explicit gate bias, bias-free SwiGLU), and the dose-response on two of them. Across an eight-model scan the mechanism is present in every GELU and SiLU family and absent only in OPT, where an opposing LayerNorm bias cancels the carried reference. Gate thresholds are implemented as couplings to a constant the network builds for the distribution it is reading (part carried in the stream and part in the parameters, in a proportion the architecture sets), rather than as bias parameters.

1 Introduction

Transformer residual streams contain large, stable components that are shared across inputs: attention sinks and massive activations (Xiao et al., 2023; Sun et al., 2024), and outlier dimensions tied to token frequency (Kovaleva et al., 2021; Timkey and van Schijndel, 2021; Puccetti et al., 2022). Existing work locates these objects and shows they matter for quantisation and attention. What they are *for*, computationally, is less settled.

This paper studies one such object, the corpus-mean direction of the layer-normalised residual stream, b_L , and measures a specific function for it: it is the reference against which the MLP gate population sets its operating point. The claim decomposes into parts that were tested separately, each against a random-initialised control model (“twin”) of identical architecture:

1. **Object** (§2): b_L is constant across text, is nearly a single direction shared across mid-stack layers, and its mid-stack content is not the surface function-word distribution.

2. **Function** (§3): removing or amplifying the stream’s projection on b_L releases or suppresses gate firing, dose-monotonically, far beyond norm-matched controls; the twin is flat.
3. **Identity** (§4): the function is carried by the direction, not by magnitude.
4. **Implementation** (§5): resting gate inhibition is the coupling $w \cdot b_L$, not the bias parameter. This is an exact decomposition, not a fit.
5. **Persistence and dependence** (§6): the direction is continuously re-supplied by aligned per-layer mean writes, so it re-accumulates after local removal; defeating that restoration by co-removal is near-catastrophic for prediction.
6. **Development** (§7): the operating point is built early — on Pythia-410M it forms in the induction-transition window; a finer 160M grid shows it forms first; and it forms even with attention disabled, so it is set by the token-local prediction task.

The battery is characterised in Phi-2 (Abdin et al., 2023), with Development on Pythia-410M (Biderman et al., 2023) checkpoints and two small trained models. The two most portable legs, the static decomposition (§5) and the causal dose-response (§3), are then tested on further families; the remaining causal tests are Phi-2-only (Limitations).

Throughout, “gate” means the scalar nonlinearity of an MLP hidden unit; “duty cycle” is the fraction of positions at which a unit’s activation is positive. Two background facts motivate the hypothesis. First, the trained model’s median duty cycle is 0.076 against 0.48 at random init: trained gates are off by default (Szatkowski et al., 2025), and “off by default” is only defined relative to what occupies the stream at rest. Second, in a decomposition of what drives firing, the corpus-mean component contributes a share of about -0.40 of trained drive at every layer measured, and ≈ 0 in the twin: trained gates are held down by something constant. The experiments below ask whether that constant is functional structure.

Provenance. Every experiment specified its outcome branches in advance, with a construction-validity gate on each instrument (it must first reproduce a known result before its new reading is taken). Primary model: Phi-2 (Abdin et al., 2023) (2.7B, 32 layers, $d=2560$), fp16, hooks on public weights; twin: same architecture, random init. Cross-model replication: GPT-2-medium (Radford et al., 2019), Pythia-1.4B (Biderman et al., 2023), Qwen2.5-1.5B (Qwen et al., 2024), and TinyLlama-1.1B (Zhang et al., 2024) carry the causal and static legs; the static decomposition also runs across an eight-model GELU/ReLU/SiLU scan comprising those four plus Pythia-410M, SmoLM2 (Ben Allal et al., 2025), and two OPT models (opt-350m, opt-1.3b) (Zhang et al., 2022). Each is compared against its own `from_config` twin. Development (§7): Pythia-410M-deduped public checkpoints (Biderman et al., 2023), plus a 160M model (Pythia-160M architecture, standard GELU MLP) trained here on the Pile with dense early checkpoints, and an attention-disabled variant of it (each layer’s attention output projection zeroed and frozen).

2 The object

Definitions. m_L is the corpus-mean raw residual at the input of layer L ; b_L is the corpus-mean layer-normalised MLP input at L (the object gates actually read); $\hat{m}_L$ the unit vector of m_L . Statistics are computed over document positions ≥ 16 , excluding the attention-sink region.

Constancy. Split-half cosine of b_L across document halves is 0.94–0.99 at every measured layer.

A constant fraction, not a constant magnitude. The residual stream grows several-fold across the stack, so a fixed coupling could not be a fixed operating point unless something normalised it. It does. Over L2–L28, mean $\|x\|$ at the layer input grows $25.9 \rightarrow 131.9$ ($5.1\times$) and $\|m_L\|$ grows $8.7 \rightarrow 51.1$ ($5.9\times$), so the background’s *share* of the stream stays in 0.33–0.44 throughout: the pre-norm mean grows only to track the stream it rides. LayerNorm then divides that growth out, and the object gates read is flat, $\|b_L\| = 6.6\text{--}10.1$ over the same span. The resting term of §5 inherits the stability directly (median -1.57 at L6 to -1.19 at L28, within a band of about 0.7 nats while the stream quintuples). The twin separates this as a trained property: there the share is not stabilised at all but climbs monotonically ($0.54 \rightarrow 0.78$ over the same layers) and $\|b_L\|$ climbs with it ($27.2 \rightarrow 39.6$). What the network carries is therefore more precisely a maintained constant *fraction* than a constant magnitude; the pre-norm magnitude is not the invariant, the post-norm operating point is.

What holds the fraction. The stability above is asserted by the split-half and depth measurements but not explained by them, and it has an exact decomposition. Writing $x_{\text{in}} = b_L + \delta$, the reference component is $\langle x_{\text{in}}, \hat{b}_L \rangle = \|b_L\| + \langle \delta, \hat{b}_L \rangle$, and because b_L is the corpus mean the second term has mean zero by construction: only its variance is at issue. Two things suppress it. First, LayerNorm’s per-position denominator: recomputing every contribution with σ frozen at its corpus mean raises the variance of the reference component from 0.62/0.51/0.96 to 4.25/2.28/2.83 at L8/L16/L24, so the per-position gain control removes 85%/78%/66% of it. Second, aggregation: the remainder is a sum over the 17–49 writes upstream of the layer, none contributing more than 0.33 of standard deviation against a total mean near 6.7, and the sum is far steadier than its parts (coefficient of variation 0.118 against 0.25–6.65 for individual sources). The sources do *not* cancel: their signed contributions reinforce, with mean pairwise correlation $+0.05$ and covariance adding 34–72% of the total variance, so this is normalisation plus averaging, not a balance. Two by-products of the same decomposition: the largest and steadiest single contributor is the layer-0 block (its attention and MLP writes supply 28% of the reference component at L8, with mean-to-spread ratios of 3.1 and 4.0 against 1.3–1.5 for late writes), and aggregation *de-correlates* as well as stabilises: a held-out ridge on twelve surface text features predicts the largest individual contribution at $R^2 = 0.34$ but the total at only 0.11 (permutation null -0.001), which is the behaviour a reference needs, since a yardstick that tracked token frequency would not be one.

A slowly rotating band, not one direction. $\cos(\hat{m}_{10}, \hat{m}_{18}) = 0.94$, and the reference gates actually read is aligned over the same span ($\cos(\hat{b}_{L10}, \hat{b}_{L18}) = 0.92$). Extending the comparison to every layer shows a band rather than a single vector: adjacent layers sit at ≈ 0.97 , but $\cos(\hat{b}_{L6}, \hat{b}_{L22}) = 0.81$ and $\cos(\hat{b}_{L6}, \hat{b}_{L30}) = 0.24$. What survives the drift is the coupling rather than the direction. Substituting another layer’s reference into layer L ’s resting decomposition, at layer L ’s own magnitude, still orders layer L ’s duty at $\rho = 0.86\text{--}0.96$ anywhere mid-stack and at $0.69\text{--}0.77$ even from L30, against $0.06\text{--}0.09$ for a random direction (§5); in the twin the same transfer instead tracks the cosine almost exactly ($0.44 \rightarrow 0.41$, $0.71 \rightarrow 0.70$), so the tolerance is trained. Causally the same point is made in §4: layer 10’s gates accept layer 18’s mean direction as their reference almost without loss. This cross-layer geometry is a Phi-2 property, not a universal one (it is family-dependent across the replication models; Limitations); every other claim in the paper is per-layer and stands without it.

Content. Decoded through the *final* unembedding, mid-stack b_L (L2–L18) aligns only weakly with any single token direction (max row cosine $0.14\text{--}0.15$, about $1.7\times$ the isotropic floor), rising toward the readout. This paragraph is descriptive; no claim in §3–§7 depends on it.

α	$\Delta\text{frac}(\text{act}>0.5)$			KL vs. clean		
	L6	L10	L14	L6	L10	L14
0.0 (removed)	+0.103	+0.086	+0.075	1.427	0.519	0.274
0.5	+0.023	+0.023	+0.021	0.047	0.040	0.035
1.0 (identity)	0.000	0.000	0.000	0.000	0.000	0.000
1.5	−0.006	−0.007	−0.007	0.014	0.018	0.020
2.0 (doubled)	−0.008	−0.009	−0.009	0.055	0.060	0.110
norm-matched control	+0.003	+0.001	+0.003	0.020	0.022	0.025

Table 1: Dose-response of gate firing and output damage to scaling the stream’s projection on $\hat{m}_L$ at the manipulated layer (Phi-2, trained). The twin is flat at every α ($|\Delta\text{frac}| \leq 0.002$, no ordering). Point estimates are document medians over 39 documents; 95% document-bootstrap CIs (2000 re-samples) on the removal row: Δfrac $[+0.096, +0.112]/[+0.079, +0.098]/[+0.068, +0.088]$ and KL $[1.14, 2.03]/[0.36, 0.75]/[0.15, 0.40]$ (L6/L10/L14), each disjoint from the norm-matched control’s KL CI ($\leq [0.010, 0.032]$). The document-sampling spread on KL dwarfs the run-to-run fp16 jitter ($\sim 1\%$), so only the leading two digits are informative.

Sources. The embedding mean is negligible (norm 0.42, cosine 0.02 to the deep mean). Instead, every layer’s attention *and* MLP mean writes are mutually aligned with the deep mean (cosines $+0.2$ to $+0.7$, never negative), each small (norms ~ 1 – 3 against update RMS 15–38), summing coherently: the reference is a distributed construction to which every component contributes a small aligned DC term.

3 Function: the gate population thresholds against it

Manipulation. At layer $L \in \{6, 10, 14\}$, the residual entering the layer is modified at all positions ≥ 16 : $x' = x + (\alpha - 1)(x \cdot \hat{m}_L) \hat{m}_L$, for $\alpha \in \{0, 0.5, 1, 1.5, 2\}$. $\alpha=1$ is the identity (verified as an exact no-op, KL = 0); $\alpha=0$ removes the stream’s projection on the reference; $\alpha=2$ doubles it. The control subtracts the same per-position magnitude along a fixed random unit direction (norm-matched). 39 documents.

Result. Table 1 shows the dose-response. Removal releases firing; amplification suppresses it; the response is monotone through the identity point at every layer and 30 – $40\times$ the norm-matched control, consistent with a population of near-binary gates driven across threshold together (Balogh, 2026). Against the clean baseline of $\text{frac}(\text{act}>0.5) = 0.0105$ at L10, the release is a $\sim 9\times$ multiplication of above-threshold firing. The suppression side saturates small, which is the expected asymmetry: at duty 0.076 the population is already near its inhibited floor, so added inhibition has little left to turn off while release frees every marginal gate. The twin shows no response, having no resting coupling to lose (§5).

Output damage. Removal at a single site costs KL = 1.43/0.52/0.27 (L6/L10/L14) against 0.020–0.025 for the norm-matched control. The twin loses only 0.32–0.38 KL under the same removal despite its raw mean being 8 – $11\times$ larger in norm (168 vs. 15–21): per unit of removed norm the trained model is roughly two orders of magnitude more sensitive.

A matched-energy scale. Two matched-energy controls place the removal cost. Isotropic noise injected at one layer carrying b_L ’s own energy share ($\sim 10\%$, §2) costs KL 0.013 at both L6 and L10, against 1.68 (L6) and 0.69 (L10) for removing b_L on the same per-position measure (a mean over positions, distinct

from Table 1’s final-position KL): the reference removal is $127\times$ (L6) to $52\times$ (L10) more damaging than matched-energy noise, and $\sim 1000\times$ more than removing a random direction. Turned around, injecting per-position noise *along* b_L costs $8\text{--}42\times$ the same energy injected along a matched random direction, across layers and amplitudes; enforcing an exactly zero empirical mean on the injection (residual $\|\langle n \rangle\|/\|b_L\| \leq 0.01$) leaves the ratio unchanged, so it is per-position jitter on the reference axis, not a residual shift of the operating point. Both reproduce the dose-response of Table 1 by an independent route, on KL rather than firing rate.

Where the damage lands. Resolving the same removal by context locates the cost. Applying it at L16 on a 16-sentence corpus costs $\Delta\text{NLL} = +0.493$ and $\Delta H = +0.379$ while keeping top-1 at 0.70, and the loss is concentrated: $+0.94$ at high-entropy positions against $+0.04$ at low-entropy ones, and $+0.88$ at function-word positions against $+0.36$ at content words. Replacing the massive-activation coordinates with their corpus constants at the same site is a much milder intervention on the same readout ($\Delta\text{NLL} = +0.086$, top-1 0.87), and its damage sits in the same places. These are the positions that most need the gate population working: removing the reference costs most at the high-entropy, least-certain predictions and little where the next token is already near-settled.

An input-side demonstration, and what supplies the reference. Every manipulation above is an activation edit, which leaves open the objection that edited states are off the distribution the model ever visits (Mishra et al., 2026). The same conclusion can be reached without editing anything. Feeding a degenerate input (one token repeated) inflates LayerNorm’s per-position denominator by $8\text{--}15\times$, which divides the reference out rather than rotating it away: the reference component falls from 6.8 to 0.9 at L8 and from 7.1 to 1.0 at L24, going slightly negative in the most degenerate condition. Because the coupling is inhibitory, removing it releases firing, and duty rises from 0.087 to 0.791 at L24 (a $9\times$ increase) with the weight-space resting term’s rank correlation to measured mean pre-activation falling from 0.98 to 0.16. This is the same release the α knob produces, obtained from a real prompt.

Two scope statements belong with it, because they bound how much the model depends on its input distribution. The failure needs extreme degeneracy: tiling the first k tokens of a document, the denominator is inflated at $k = 1$ and $k = 2$ but is already back to its prose value at $k = 4$ ($\sigma = 15.2/2.7/0.92$ against a prose 0.99), and an i.i.d. draw from four distinct token types is enough. And recovery is local: appending a clean prompt after a degenerate context restores duty within two or three tokens ($0.739 \rightarrow 0.542 \rightarrow 0.207 \rightarrow 0.186$ by offset at L24, against a prose 0.08). Natural text is therefore nowhere near the failure regime, and the reference is supplied by a short window of ordinary recent context rather than by any global property of the document. Relatedly, b_L is conditioned on the input distribution rather than intrinsic to the model: $\cos$ between the reference estimated on prose and on other inputs is 0.97–0.98 for held-out prose, 0.76–0.84 for code, and 0.02–0.05 for a repeated token, where the twin has one direction for all of them.

Restoration. The release is local *in depth*: downstream firing is undisturbed. After removal at L6, $\cos(\text{resid}, \hat{m})$ is down -0.16 at L10 and L14, -0.10 at L18, and recovers to baseline by L26–L30: the stream re-accumulates its mean over $\sim 12\text{--}16$ layers, consistent with the anatomy (every layer’s mean write is reference-aligned, so the reference self-heals by accumulation). Deeper removal sites restore more slowly. It is *not* local in sequence position, which is a separate axis and worth stating because the two are easy to conflate: removing the reference in an eight-position window at L6 and then reading only *untouched* later positions, the next-token distribution is still altered 32 tokens past the window at KL = 0.098, 16% of the in-window magnitude and $54\times$ the norm-matched control, and the ratio to control

replacement $\hat{c}$	$\cos(\hat{c}, \hat{m}_{10})$	$\Delta\text{frac}(\text{act}>0.5)$	KL
none (pure removal)	—	+0.0845	0.521
coordinate-shuffled $\hat{m}$	−0.003	+0.0840	0.963
random direction	+0.011	+0.0778	0.502
function-word W_U sum	+0.096	+0.0740	0.402
layer 18’s $\hat{m}$	+0.940	+0.0027	0.004
identity	1.000	0.000	0.000

Table 2: Replacement panel at L10 (Phi-2, 23 documents). Preservation of quiet firing tracks $\cos(\hat{c}, \hat{m})$ monotonically.

grows with distance, since the control decays faster than the effect. Greedy continuations diverge from the unperturbed ones within a median of two steps. The effect is an alteration rather than a degradation: KL is large while the change in the true token’s log-probability is inconsistent in sign, so the reference’s influence extends across tens of positions without later predictions being systematically worse.

Cross-model. The same knob at one mid-stack layer (depth ≈ 0.31 : Pythia-1.4B L8, Qwen2.5-1.5B L9; 29 documents, each against its own twin) reproduces the sign, the monotonicity, and the control separation. On the duty-cycle readout (activation > 0), removal shifts duty by +0.318 [0.303, 0.329] (Pythia) and +0.302 [0.286, 0.316] (Qwen), doubling by −0.124 and −0.081, monotone across a 15-point α grid, against +0.0001 for the norm-matched control and flat twins ($|\Delta\text{duty}| \leq 0.002$); Phi-2 on the same measure is +0.238/−0.070. Qwen is the sharpest case: its SwiGLU gate has no bias parameter, so the carried reference is the entire resting inhibition, and scaling it still moves firing in dose-response. Removal costs final-position KL 3.60 (Pythia) and 0.59 (Qwen). Weakening the reference also raises output entropy in every family (the release arm is universal; ΔH at $\alpha=0.8$ is +0.057 Pythia, +0.047 Qwen, +0.013 Phi-2, each above a norm-matched control $\leq |0.0005|$), with the minimum at the natural magnitude.

4 Identity: the direction, not the magnitude

Removal tests necessity of the projection, but it leaves two alternative readings open. Gates might not read this direction at all and simply need the stream to be full; and since the manipulation acts on the pre-norm residual, removing the projection shrinks the norm entering LayerNorm, so the renormalisation boosts the gain on every orthogonal component, so some of the release could be that gain boost rather than the loss of the inhibitory coupling. The replacement panel addresses both: at L10, $x' = x - (x \cdot \hat{m})\hat{m} + (x \cdot \hat{m})\hat{c}$ transfers the signed per-position projection onto a candidate direction $\hat{c}$; magnitude, sign pattern, and timing are preserved, the total norm is approximately unchanged (the cross-terms with $\hat{c}$ are small), and only the direction changes.

In Table 2, shuffled and random directions release firing exactly like removal; a direction at cosine 0.096 to $\hat{m}$ buys a sliver proportionate to its cosine and nothing more; and another layer’s mean direction (cosine 0.94) preserves quiet firing almost perfectly, $31\times$ less release and $130\times$ less KL than removal. The magnitude-reader alternative is excluded: no low-cosine candidate preserves anything. The gain-boost alternative is excluded the same way: a random-direction replacement leaves the pre-norm magnitude in place, so the LayerNorm gain is the same as clean, yet firing releases as if the projection had been removed, while layer 18’s direction, under the same norm, preserves quiet firing. One further observation:

layer	$w \cdot b_L$ (median)	b_n (median)	resting	ref-dominant	$\rho(\text{resting, duty})$	$\rho(\text{random dir})$
6	-1.535	-0.027	-1.568	99.98%	0.943	0.058
10	-1.226	-0.022	-1.250	99.99%	0.959	0.072
14	-1.091	-0.021	-1.115	99.96%	0.960	0.085
22	-1.130	-0.018	-1.149	99.94%	0.954	0.073

Table 3: Exact resting decomposition of gate pre-activations (Phi-2, 19 documents). “Ref-dominant” is the fraction of gates whose resting inhibition is carried mostly by $w \cdot b_L$. The last column repeats the rank correlation with a random direction substituted for b_L at matched norm (median of 50 draws, $p_{95} \leq 0.192$, maximum 0.223; from a 39-document run, which reproduces the $\rho(\text{resting, duty})$ column at 0.946–0.964). The $\rho(\text{resting, duty})$ column is near-tautological read alone; it is the *contrast* between the last two columns that carries direction specificity. 95% document-bootstrap CIs (2000 resamples): ref-dominant stays $\geq 99.89\%$ at every layer (widest [99.89, 99.98] at L22), and ρ is [0.936, 0.945]/[0.951, 0.959]/[0.953, 0.959]/[0.945, 0.952] (L6/L10/L14/L22). The *resting* column is a per-gate median and does not equal the sum of the $w \cdot b_L$ and b_n medians, since a median does not distribute over addition across a heterogeneous gate population.

the shuffled replacement’s KL (0.96) *exceeds* removal’s (0.52); transferring the projection onto a wrong direction injects a coherent error signal on top of losing the reference.

5 Implementation: the threshold is the coupling

A gate’s *resting* pre-activation decomposes exactly: $\langle w \cdot x_{\text{in}} + b_n \rangle = w \cdot b_L + b_n$, since $\langle x_{\text{in}} \rangle = b_L$ by definition. If gates threshold against the carried reference, training should have moved the operating-point inhibition into the coupling $w \cdot b_L$ rather than the explicit bias parameter b_n .

Table 3: the reference coupling carries the resting inhibition at 50–60 $\times$ the explicit bias parameter for more than 99.9% of gates at every measured layer, and the resting pre-activation rank-predicts each neuron’s duty cycle at $\rho \approx 0.95$. The second of those figures needs its null before it can be read, and the next paragraph supplies one; the claim it survives to support is the narrower and more useful one, that the population’s operating point is set along *this* direction rather than by the bias parameter. In the twin, $w \cdot b_L \approx 0$ (median magnitude ≤ 0.02) and duty is 0.5; its bias parameters are exactly zero at init, so the informative twin statistic is the absent coupling. This is the classical bias-as-weight-to-a-constant-input construction, discovered in the weights: the model implements thresholds as couplings to a constant it itself carries. It also explains the dose-response of §3 without further assumptions: scaling the stream’s projection on $\hat{m}$ scales the operating-point term of every gate simultaneously.

What the duty correlation does and does not show. Since $b_L := \langle x_{\text{in}} \rangle$, the resting term *is* the mean pre-activation, and duty is the fraction of that same distribution above zero. A rank correlation between the mean of a distribution and $P(> 0)$ is high for any family with comparable spreads, so $\rho \approx 0.95$ may be a property of the construction rather than of b_L , and the twin cannot adjudicate: with resting ≈ 0 for every gate it has no spread to correlate. We therefore built the null the twin does not provide. Permuting the resting values across neurons (each gate keeps its own centred pre-activation distribution but receives an offset carrying no information about b_L) still yields $\rho = 0.87$ –0.93 (L6/L10/L14/L22), which is 92–97% of the measured value. Injecting offsets drawn from the trained resting distribution into the *twin*, a model with no coupling at all, yields $\rho = 0.984$, higher than the trained model, because the statistic is governed by how homogeneous the per-gate spreads are (twin coefficient of variation 0.056 against the

trained model’s 0.15–0.20) rather than by the reference. Consistently, $\rho(\text{resting}/\sigma_n, \text{duty}) = 0.987\text{--}0.994$ exceeds the raw figure. The duty correlation is a location fact and we do not rest anything on it.

Two statistics do survive that test. First, direction specificity: a random direction at the same norm as b_L , substituted into the same decomposition, orders duty at $\rho = 0.058\text{--}0.085$ (median over 50 draws; $p_{95} \leq 0.192$, maximum 0.223) against the reference’s 0.946–0.964. The operating point is set along this direction and not along an arbitrary one, which is the static counterpart of the replacement panel of §4. Second, the coupling is largely norm-scaled: the read-weight norm alone orders duty at $\rho = -0.878$ to -0.896 , as the narrow $|c_n|/\|w\|$ band below implies, and in the twin the same statistic is null (-0.010 to $+0.007$), so the norm-to-duty relation is itself trained. The decomposition of the operating point is a norm-scaled resting depth acting along a specific carried direction; the residue above the norm-only ordering, not the 0.95, is what b_L contributes to the ranking.

Per-gate anatomy. Factoring each fc1 row against the unit reference, $w = c_n \hat{b}_L + w_\perp$, the coupling is a small slice of read norm: $|c_n|/\|w\|$ has median 0.07–0.12 across layers against a random-twin floor of 0.013, thinning toward the readout, so ≈ 0.99 of the row remains free for content. The coupling alone rank-predicts duty as well as the full resting inhibition ($\rho(c_n, \text{duty}) = 0.943\text{--}0.959$, matching Table 3), so b_n adds nothing to the ordering; both figures inherit the caveat above. b_n is redundant here rather than uninformative: on its own it orders duty at $\rho = 0.81$. Per gate, the coupling is a low single-digit-percent tax on the read weight that fixes where the gate sits while leaving its content direction almost untouched: $c_n = w \cdot \hat{b}_L$ is at once the gate’s (negative) read of the reference and the term that sets its resting operating point, so the read and write sides are one coupling.

The coupling over training. The decomposition above compares two endpoints. Pythia-410M-deduped, whose GELU gates carry an explicit bias and whose checkpoints are public, lets the two static terms be tracked step by step (four mid-stack layers). b_n is exactly zero at initialisation (Pythia zero-inits biases) and stays ≤ 0.04 in magnitude the entire run, while $|w \cdot b_L|$, after training first destroys the init constant ($\sim 0.25 \rightarrow 0.12$ by step 128), builds through the step 512 $\rightarrow$ 1000 window ($0.16 \rightarrow 0.38$), peaks near 0.85–0.95 around step 4000, and relaxes partially thereafter. The reference coupling carries the resting inhibition of $\geq 99.6\%$ of gates at every checkpoint including init: there is no phase in which the explicit bias leads and is overtaken. The network does not route the operating point through the bias parameter at any stage; it builds the coupling directly. Duty falls $0.51 \rightarrow 0.16$ over the same build and $\rho(\text{resting}, \text{duty})$ stays 0.91–0.998 throughout. The endpoint inference is an observed time-course, and it independently reproduces the onset window of §7 through a different statistic.

Cross-model replication. The resting decomposition is static (weights and mean activations only), so it can be re-run cheaply on other families against a `from_config` twin. Table 4 does so on four more architectures, spanning both gate types: GELU MLPs with an explicit gate bias (GPT-2-medium, Pythia-1.4B) and bias-free SwiGLU gates (Qwen2.5-1.5B, TinyLlama-1.1B). In every family the resting pre-activation is negative (gates held off by the carried reference) and rank-predicts duty at $\rho \geq 0.96$ across mid-stack layers, with the twin floored (resting ≈ 0 , duty ≈ 0.5) throughout. Every ρ in this paragraph and in Table 4 inherits the caveat above and is reported as description; what carries the cross-family claim is the sign and magnitude of the resting term relative to b_n , and the random-direction control was run on Phi-2 only. Where a gate bias exists, the reference coupling dominates it for $> 99\%$ of gates at $15\text{--}43\times$ the bias parameter; where the gate is bias-free, the coupling alone carries the entire negative resting term. The one sub-threshold ρ is Qwen’s shallowest sampled layer ($L5$, $\rho = 0.62$, a strongly suppressed near-off

model	gate (bias?)	resting med.	ref-dom.	$\rho(\text{rest, duty})$
GPT-2-medium	c_fc (yes)	−1.52 to −0.99	99.4–99.7%	0.91–0.97
Pythia-1.4B	h_to_4h (yes)	−1.08 to −0.74	> 99.9%	0.96–0.98
Qwen2.5-1.5B	gate_proj (no)	−4.14 to −0.37	—	0.62–0.99
TinyLlama-1.1B	gate_proj (no)	−0.21 to −0.07	—	0.97–0.99

Table 4: Cross-model resting decomposition (30 documents, four mid-stack layers per model at depth fractions 0.2–0.65; ranges span those layers). “Ref-dom.” is the fraction of gates whose resting inhibition is carried mostly by $w \cdot b_L$ rather than the bias parameter (vacuous for the bias-free SwiGLU gates; for these the informative statistic is the negative resting coupling and its duty-prediction). Where a bias exists, $|w \cdot b_L|/|b_n|$ is 15–18 $\times$ (GPT-2) and 38–43 $\times$ (Pythia). Every model’s random-init twin is floored: resting median $|\leq 0.01|$, duty ≈ 0.5 .

regime, duty 0.004); every deeper layer is $\rho \geq 0.96$. Magnitude varies (TinyLlama’s resting inhibition is an order of magnitude weaker than GPT-2’s).

A counteracting boundary case. The mechanism is not universal. Extending the static decomposition across the eight-model GELU/ReLU/SiLU scan, it is present in all six non-OPT families (ref-dominant 99.5–100% where a bias exists, $\rho \geq 0.89$ throughout) but absent in both OPT models (opt-350m, opt-1.3b): their resting inhibition is 40–130 $\times$ weaker, bias-dominated, and rank-predicts nothing ($\rho = 0.000$, as follows from a resting term near zero). A norm decomposition explains why. Writing $w \cdot b_L = w \cdot (b_L - \beta) + w \cdot \beta$ (residual-carried mean plus LayerNorm bias), the residual-carried part is primary wherever the mechanism is present: it dominates 91–99% of gates in the GELU LayerNorm models and equals the resting term by construction in the bias-free RMSNorm models, with β a 7–26% augmentation. So the §6 attribution of the reference to a residual-carried mean, rather than a static norm parameter, holds. OPT is the exception that locates the requirement: it builds a large residual-carried inhibition comparable to GPT-2’s but trains an almost exactly opposing LN-bias excitation ($|w \cdot (b_L - \beta)| \approx |w \cdot \beta| \approx 0.5$ –0.6, opposite sign), netting $w \cdot b_L \approx 0.006$: a reference cancelled at the operating point, not absent. ReLU-versus-family stays formally confounded (every ReLU model sampled is an OPT).

How much of the reference is a parameter. The decomposition above separates the residual-carried mean from the LayerNorm bias, but it counts the accumulated *module* biases (each block’s attention-output and MLP-output bias) inside the residual-carried part, though they are input-independent, added unconditionally whatever the input. Separating them changes the accounting materially. Projecting β and every upstream module bias onto the reference (the module biases enter as a constant vector divided by the per-position σ , so the constant is computable from weights alone), the input-independent share of the reference component tracks architecture exactly across the replication set: 41–47% (Phi-2), 10–32% (GPT-2-medium), 52–84% (Pythia-410M), 26–40% (Pythia-1.4B), and *exactly* 0% in all three bias-free RMSNorm models (Qwen2.5-1.5B, TinyLlama, SmoILM2), which have neither β nor output biases and must therefore assemble the same reference entirely from computed writes. OPT is again the informative case: its share is *negative* (−17 to −8%), the same cancellation seen in the paragraph above from the parameter side. So the mechanism is architecture-independent while its *ingredients* are not: how much of the reference is carried in the stream versus in the parameters is a proportion the architecture sets, and the span is wide, so no single figure should be quoted for it.

6 How the reference persists, and the cost of losing it

Attention carries the steady feed. Decomposing each block’s output into its component along $\hat{m}$ and the orthogonal remainder, and measuring the constant (split-half) share of each: at L10, the attention block’s $\hat{m}$ -channel has constant share 0.452 against 0.014 for its orthogonal channel, a $33\times$ dissociation; the block outputs a steady reference feed while its content channel fluctuates. The MLP shows no such dissociation (0.003 vs. 0.003): unlike attention it carries no steady reference feed, though its time-averaged write is reference-aligned (§2). In the twin both channels are largely constant (0.93 vs. 0.87; an untrained block is a mostly-constant filter), so the dissociation is a trained property. Together with §2: per-layer mean writes build the reference; attention sustains it in operation.

Co-removal. Single-site removal understates the dependence because the stream heals. Removing the projection at *all* measured layers ≥ 6 simultaneously releases firing at every depth (at L18 more than at the shallowest site) and costs $\text{KL} = 4.38$, $\Delta\text{NLL} = +4.75$ nats, about $8.5\times$ the L10 single-site removal (Table 1, KL 0.52), while the twin under the identical intervention loses 0.65 KL with flat firing (document-bootstrap 95% CIs over 23 documents: KL [3.74, 6.05], ΔNLL [4.62, 5.03], disjoint from the twin). Late-only co-removal (≥ 14) is far milder (KL 0.55, CI [0.24, 0.90]): the dependence is concentrated mid-stack. The reference is a standing operating condition of the trained trunk, not a local convenience of any layer.

7 Development

The operating point of §5 is built early in training, and by a token-local pressure. Three developmental settings, at increasing resolution, show this.

Pythia-410M: the reference forms in the induction window. On Pythia-410M-deduped public checkpoints (a different family and scale from the causal experiments; one seed), the reference coupling forms in the same early window as gate sparsification and the induction transition. In the step 512→1000 interval the induction-head score rises $0.03 \rightarrow 0.88$ (Olsson et al., 2022), the median duty cycle falls $0.42 \rightarrow 0.28$ (continuing to 0.16 by step 4000), and the reference share of resting drive deepens $-0.06 \rightarrow -0.34$ (to ≈ -0.75 by step 3000–4000). Pythia has no checkpoint between 512 and 1000, so this grid places the onset in that one interval but cannot order the three within it; below it nothing is yet moving (across log-spaced steps 1–256 the induction score is at its floor, duty ≈ 0.50 , and the reference share ≈ 0 at mid-stack), the only earlier event being training’s first act on the MLP write, which destroys the init constant (constant share $0.40 \rightarrow 0.15$ by step 128) (Gallego-Feliciano et al., 2025). The trained reference is therefore a construction, not a residue of initialisation, and it is in place by the time the induction transition — itself a multi-phase event (Minegishi et al., 2025; Xu, 2026) — completes.

A 160M model: the operating point forms first. A finer developmental grid resolves the order the Pythia checkpoints cannot. In a 160M model (Pythia-160M architecture, a standard GELU MLP, trained here on the Pile with dense early checkpoints), the gate operating point forms well before the induction transition: median mid-stack duty falls from 0.50 to 0.075 by step ≈ 800 (and lower thereafter), and the resting depth — reported here as z , the resting pre-activation divided by its per-unit standard deviation, so the operating point is in units of the gate’s own drive spread rather than the raw nats of §5 — reaches ≈ 1.5 over the same window, while the induction-head score is still near its floor and completes its rise only over steps ≈ 3000 –12000. Throughout, the operating point is the reference coupling — the mean-input

term carries ≈ 0.99 of the resting pre-activation, as in §5 — so it is the mechanism of §5 that is in place first. (Absolute step counts are not comparable to Pythia-410M’s, which uses a different schedule and $\sim 100\times$ more tokens per step; the developmental *order* is the point.)

An attention-free model: the operating point is token-local. The operating point forms without any cross-token computation. Training the same 160M architecture with attention disabled — every layer’s attention output projection zeroed and frozen, leaving a purely position-wise model that predicts each token from the current token alone, a bigram map — reproduces the same development. In a matched pair of runs differing only in whether attention is enabled (identical script and schedule, to step 3000), the attention-free model’s mid-stack duty settles at 0.077 against 0.067 for the attention-carrying one (a $\approx 15\%$ higher floor) and its operating point is slightly shallower ($z \approx 1.3$ against ≈ 1.5), while the reference coupling is unchanged (mean-input share ≈ 0.99 in both). Attention makes sparsification marginally faster and deeper, but the operating point is set by the token-local next-token prediction task and forms whether or not the model can read other positions.

Reading. Across the three settings the operating point is an early, token-local component of a trained MLP: the network sets its gates against the carried reference as it learns the base next-token map, in the same early window in which cross-token circuits mature at scale. Activation sparsity is a known endpoint property (Szatkowski et al., 2025); the reference coupling is how that endpoint is reached, and the pressure that reaches it is present even in a model with no context to read.

8 Related work

Massive activations, sinks, outlier dimensions. Attention sinks (Xiao et al., 2023) and massive activations (Sun et al., 2024) are large constant components with a bias-like role in attention; Sun et al. (2024) explicitly propose them as bias terms (see also Owen et al., 2025), and later work ties sinks to compression valleys (Queipo-de Llano et al., 2025), to outlier-driven rescaling essential for training (Qiu et al., 2026), and to depth-transient damping (Hengyu, 2026). Most recently, Sun et al. (2026) give a functional split: massive activations as near-constant hidden components acting as implicit parameters, and attention sinks as local attention modulators; they identify the pre-norm configuration as what couples the two, decoupling them under ablation. Outlier dimensions are frequency-driven and normalisation-amplified (Kovaleva et al., 2021; Timkey and van Schijndel, 2021; Puccetti et al., 2022), and Macocco et al. (2025) show that last-layer outlier dimensions act as a constant positive logit contribution toward frequent tokens. The object studied here is in the same family but the measured function differs in site and form: a *direction* (not a coordinate set), read at the MLP gate inputs mid-stack (not at the output logits), that carries the resting inhibition of the gate population by exact decomposition, is direction-specific under replacement, and is assembled by distributed aligned writes. The context-resolved removal of §3 does place the two objects in overlapping *functional* territory, since both end up supporting a default distribution over frequent tokens; we take the separation to be one of object and site rather than of function, and the geometric comparisons below are what carry it.

We ran the geometric comparison against these objects on Phi-2. Mid-stack, $\cos(b_L, \cdot)$ against the first-token direction at its emergence layer (located here at layer 21, peak coordinate magnitude ~ 1250 ; cf. Shi et al., 2026, who propose the first-token massive activation as a “shared global reference vector” for attention) is 0.04–0.17, and against the residual-sink and massive-coordinate directions ≤ 0.27 : the objects are distinct. The separation is trained: in the random-init twin the reference *is* the sink blob ($\cos = 0.98$, no massive activation), and training rotates the reference off the sink. On outlier dimensions:

b_L carries 10–23% of its squared norm on the twenty largest last-layer outlier coordinates against an isotropic share of 0.8%, a real but minority overlap that grows toward the readout; its own dominant coordinates are largely disjoint from the outlier set (top-20 Jaccard 0.05–0.14). Mid-stack, b_L is a distributed direction with a minority frequency-correlated component, not the axis-aligned outlier set, and it should not be conflated with the sink objects of Qiu et al. (2026) or Shi et al. (2026).

Distinctness under a variance-aware metric. The comparisons above use Euclidean cosines, and Ying et al. (2026) note these are unreliable when variance concentrates on a few coordinates: equal weighting lets the many low-variance dimensions inject noise that can mask genuine alignment along the loud ones. That objection is pointed here, because the objects we compare against — the residual sink and the massive activations — *are* those loud coordinates, so an equal-weight cosine could be understating an alignment with them. We therefore recomputed the comparisons under variance-aware metrics, on the same directions and corpus, so that any change is attributable to the metric. In the space b_L is defined in and gates read (the covariance of the layer-normalised MLP input), the covariance-weighted cosine of Ying et al. (2026) places b_L against the sink, first-token and massive directions at -0.23 to $+0.01$, against an empirical random-pair null of 0.03 median and 0.11 at the 95th percentile; restricting to the top-512 principal subspace gives -0.41 to -0.14 . The positive control $\cos(b_L, m_L)$ is held at 0.91–0.95 throughout. Under the metric the objection prescribes, the objects are not merely unaligned but mildly opposed. The instrument is validated on the twin, where b_L is the sink blob: there every metric returns 0.83–0.99. One trend is worth stating explicitly, since both metrics show it: the overlap with the sink grows toward the readout ($0.12 \rightarrow 0.27$ Euclidean over L6–L18, $0.08 \rightarrow 0.33$ in the principal-subspace metric), in step with the outlier-coordinate mass reported above.

Bias as coupling. Treating bias as a weight onto a constant input is the classical homogeneous-coordinates construction. Weight-based taxonomies of gated neurons (Gerstner and Schütze, 2025) and readings of MLP units as binary routers (Balogh, 2026) characterise gate function from the parameters, and massive activations have been read as carried constants with a bias-like role (Sun et al., 2024, 2026). The observation here joins these: a trained transformer implements the homogeneous-coordinates construction spontaneously, with the constant supplied by the network’s own carried mean rather than by a dedicated input. Sun et al. (2024) also report the trained converse on the attention side: augmenting self-attention with explicit learnable bias parameters yields models that do not form massive activations: given the parameter, the network uses it instead of building a carried constant. OPT plays the same role on the gate side here (§5): the parameter-based strategy exists, and most families do not take it. Finally, modern architectures dropped dense biases for training stability with no measured quality loss (Chowdhery et al., 2022); the decomposition here offers an account of why the removal is functionally free on the gate side, namely that the operating point never lived in b_n .

9 Limitations

- **Replication scope.** The static decomposition (§5) is shown on the eight-model scan and is absent in OPT, where an opposing LayerNorm bias cancels the carried reference; the ReLU-versus-OPT-family confound (every sampled ReLU model is an OPT) is not separated by matched-nonlinearity twins. The causal dose-response and the entropy release arm replicate on two further families (Pythia-1.4B, Qwen2.5-1.5B); the replacement panel (§4, one site, L10) and co-removal (§6) are Phi-2-only. The development results (§7) are trend-grade and single-seed: the Pythia-410M checkpoint time-course is one seed (15–30 documents), and the finer-grid and attention-disabled results are

each one 160M model trained here, one seed per arm. Two further results are single-model and modest- n : the depth profile of the background share (§2) and the context-resolved removal (§3, one site, 16 sentences).

- **Cross-layer geometry is family-dependent.** The rotating-band description (§2), and the transfer measurement that goes with it, are Phi-2 only; the cross-layer cosine is heterogeneous across the replication families: $\cos(\hat{m})$ between mid-stack layers is high in GPT-2-medium (0.95) and Qwen2.5-1.5B (0.96), modest in TinyLlama-1.1B (0.74), and below its own twin floor in Pythia-1.4B (0.41 vs. twin 0.67). The coupling mechanism replicates in every family; the single-carried-direction geometry does not, and is not claimed to.
- **Which statistics carry which claim.** The rank correlation between the resting term and duty is near-tautological on its own (§5), because the resting term is the mean pre-activation and duty is $P(> 0)$ of the same distribution; we report its nulls and rest nothing on it. What the static decomposition establishes is the sign and magnitude of the resting term against b_n , and, on Phi-2, direction specificity against a matched-norm random control. The functional claims rest on the causal legs (§3, §4, §6), which do not use the correlation at all. The duty ordering is also largely norm-scaled ($\rho = -0.89$ from the read norm alone), so the reference’s contribution to the ranking is the residue above that, which we have not decomposed further.
- **Within-layer atomicity.** b_L is a corpus mean, rank-1 by construction, and does hide register-conditioned secondary structure: structured, register-separable components survive removal of the $\hat{m}$ projection, measurable above a split-half floor and partly trained. But that structure carries only 2–4% of the variance around the mean, so the rank-1 treatment is a good approximation for the operating-point claims, and every causal result here concerns only the rank-1 mean.
- **Operating-point asymmetry.** The entropy response to weakening the reference (§3) rests mainly on the release direction, since the amplification arm is shallow in Phi-2 and varies across families.
- **Manipulation scope.** The reference is the corpus mean; a position-resolved background is a refinement not tested. Manipulations leave positions < 16 untouched, so attention re-import from the sink region is possible. α -scaling changes the LayerNorm input distribution, so all downstream effects pass through LN renormalisation; the norm-matched control shares this property, and the norm-preserving replacement panel (§4) excludes a pure gain-boost account at L10, the one site where it was run. The function-word replacement is one naive construction of a content-direction control, not an optimised one.
- **Geometric comparisons are activation-proxy.** The comparisons against first-token massive activations, the residual sink, and last-layer outlier dimensions (Related work) use activation-based proxies of each object on one model; a weights-only unification against each paper’s exact construction is not attempted. Two caveats attach to the variance-aware recomputation reported alongside them. Weighting instead by the *raw residual* covariance is uninformative here, because the comparison objects are themselves the high-variance coordinates: that metric returns the same value for all three proxies to three decimals, and its own random-pair null reaches 0.52–0.73 at the 95th percentile, so it cannot discriminate. And an inverse-covariance (whitening) variant fails its own positive control at this sample size (9,360 positions for a 2560-dimensional covariance), ranking $\cos(b_L, \text{sink})$ above $\cos(b_L, m_L)$; we report it as excluded rather than as evidence. The recomputation is Phi-2-only, like the comparisons it checks.

10 Reproducibility

Scripts and JSON artifacts (public checkpoints, fixed seeds):

`census_background_reference.py` (§2, §3), `census_background_identity.py` (§4, co-removal), `census_reference_mechanism.py` (§5, §6), `census_checkpoint_trajectory.py`, `census_formation_microscopy.py` and `census_formation_prelude.py` (§7, Pythia-410M; the last is the sub-512 resolution check), `induction_gate_coreg.py` (§7, the 160M finer-grid duty-vs-induction time-course) and `train_noattn_sparsification.py` (§7, the 160M model and its attention-disabled variant), `census_reference_geometry.py` (geometric comparisons, Related work), `census_reference_geometry_mahalanobis.py` (variance-aware recomputation of those comparisons, with per-metric empirical nulls), `census_reference_operating_point.py` (§2, the depth profile of the background share and the post-norm reference), `census_reference_statistic_nulls.py` (§2, §5: the permuted-offset and twin-injection nulls for the duty correlation, and the cross-layer cosine and transfer matrices), `census_reference_duty_controls.py` (§5, the random-direction, read-norm and bias-only controls), `atlas_backbone.py` (§3, context-resolved removal), `census_noise_floor.py` and `census_noise_zero_mean.py` (§3, the matched-energy noise floor and the direction-matched injection), `census_crossmodel_causal.py` (cross-model dose-response and entropy response), `census_bias_migration_scale.py` (Table 4), `census_bias_migration_formation.py` (coupling time-course over checkpoints), `census_bias_migration_nonlinearity.py` (eight-model scan, OPT boundary), `census_reference_norm_decomp.py` (residual-versus-LN-bias decomposition), `census_reference_corpus.py` (register adaptivity), `census_readweight_factorization.py` (per-gate anatomy), `census_reference_variance_sources.py` (what holds the fraction, §2), `census_reference_bias_and_depth.py` and `census_reference_bias_share_xmodel.py` (the parameter share, §5), `census_reference_text_statistics.py`, `census_sigma_dose_response.py` and `census_duty_at_task_position.py` (the input-side demonstration and its scope, §3), `census_reference_horizon.py` (the sequence-position horizon). Each experiment’s planned outcome branches, gates, and full numbers are in the project’s lab notebook (`plan_mid_stack_empirical_basis.md`, §3t–§3x). Code and data are available at <https://github.com/EvidentSolutions/llm-interp/tree/main/background> and archived at <https://doi.org/10.5281/zenodo.21498411>.

Use of AI Assistants

Large language models were used as assistive tools for coding, running experiments, and drafting text. All research questions, experimental design, and reported claims were directed and verified by the authors, who take full responsibility for the content.

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